

Week 4 : Contour Integration

(C+B Chapter 4)

We define the contour integral of a function $f: D \subseteq \mathbb{C} \rightarrow \mathbb{C}$ as a line integral.

Def: If C is a contour in D and $f: D \rightarrow \mathbb{C}$

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

where $z: [a, b] \rightarrow \mathbb{C}$ is a parameterization of the contour with $z(a) = z_1$, $z(b) = z_2$.

Recall that C is a contour so.

- z is piecewise continuous
- z' is piecewise continuous
- C has a finite length.

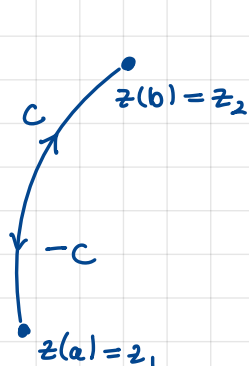
Properties:

- $\int_C \alpha f(z) dz = \alpha \int_C f(z) dz$ for all $\alpha \in \mathbb{C}$

- $\int_C f(z) + g(z) dz = \int_C f(z) dz + \int_C g(z) dz$

- $\int_{-C} f(z) dz = - \int_C f(z) dz$.

For the final property, we recall contours have orientation. Hence, $-C$ is the contour C but traversed in the opposite direction. To find a parameterization for $-C$, we then take $z(t)$ for C and run the t parameter backwards.



$$\int_{-C} f(z) dz = \int_{-b}^{-a} f(z(-t)) \frac{d}{dt}(z(-t)) dt$$

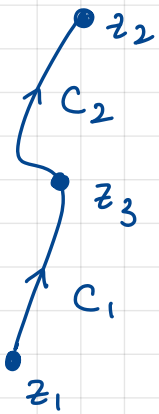
$$= \int_{-b}^{-a} -f(z(-t)) z'(-t) dt$$

$$\begin{aligned} s &= -t \\ ds &= -dt \end{aligned}$$

$$= \int_b^a f(z(s)) z'(s) ds$$

$$= - \int_a^b f(z) dz$$

We can similarly think about adding and subtracting contours.

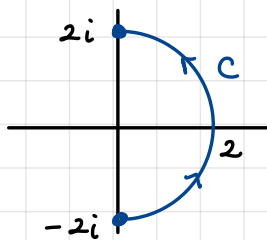


If C is a contour from z_1 to z_2 , we can break C into two pieces at z_3 . The integral then also breaks into two pieces.

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$

We will call $C = C_1 + C_2$.

Some Examples



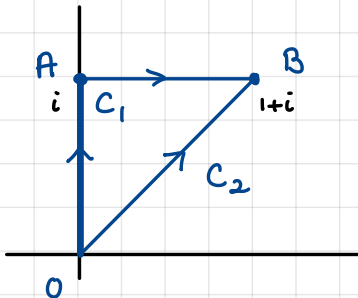
$$z = 2e^{i\theta} \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$(1) \quad \int_C \bar{z} dz = \int_{-\pi/2}^{\pi/2} \overline{2e^{i\theta}} (2e^{i\theta})' d\theta = \int_{-\pi/2}^{\pi/2} 4i d\theta = 4\pi i$$

$$(2) \quad \int_C \frac{1}{z} dz = \int_{-\pi/2}^{\pi/2} \frac{1}{2e^{i\theta}} \cdot (2e^{i\theta})' d\theta = \int_{-\pi/2}^{\pi/2} i d\theta = \pi i$$

Note that for $z \in C$ we have $\bar{z} = \frac{\bar{z}z}{z} = \frac{|z|^2}{z} = \frac{4}{z}$.

$$\text{ie.} \quad \int_C \bar{z} dz = 4 \int_C \frac{1}{z} dz$$



C_1 has parameterization:

$$z(t) = \begin{cases} ti & , t \in [0,1] \\ t+i & , t \in [0,1] \end{cases}$$

C_2 has parameterization:

$$z(t) = (1+i)t, \quad t \in [0,1]$$

$$\begin{aligned}
 (3) \quad \int_{C_1} y - x - i3x^2 dz &= \int_{0A} y - x - i3x^2 dz + \int_{AB} y - x + i3x^2 dz \\
 &= \int_0^1 t \cdot i dt + \int_0^1 [1 - t - i3t^2] \cdot 1 dt \\
 &= i \frac{1}{2} t^2 \Big|_0^1 + t - \frac{1}{2} t^2 - i t^3 \Big|_0^1 \\
 &= \frac{1}{2} i + 1 - \frac{1}{2} - i = \frac{1}{2} - \frac{1}{2} i .
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad \int_{C_2} y - x - i3x^2 dz &= \int_0^1 (t - t - i3t^2)(1+i) dt \\
 &= -(1+i) i t^3 \Big|_0^1 = 1 - i
 \end{aligned}$$

Note the integrals $\int_{C_1} y - x - i3x^2 dz \neq \int_{C_2} y - x - i3x^2 dz$.

Also

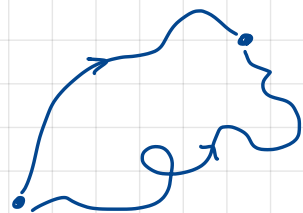
$$\int_{C_1 - C_2} y - x - i3x^2 dz = \left(\frac{1}{2} - \frac{1}{2} i \right) - (1 - i) = -\frac{1}{2} + \frac{1}{2} i .$$

(5) Suppose we want to calculate

$$\int_C z dz = \int_a^b z(t) z'(t) dt = \int_a^b \frac{d}{dt} \left(\frac{z^2(t)}{2} \right) dt = \frac{z_2^2 - z_1^2}{2}$$

That is, the value of the integral along a smooth arc only depends on the end points.

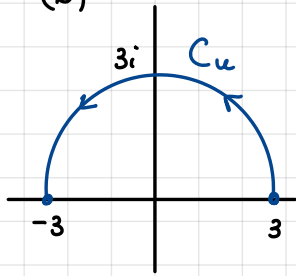
Note: $\frac{z^2}{2}$ is the anti-derivative of z .



This is true on any contour C , since C is made up of a finite number N of arcs $z_j(t)$.

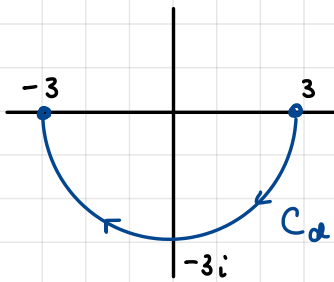
$$\int_C z dz = \sum_{j=1}^N \int_{C_j} z dz = \sum_{j=1}^N \frac{z_j^2 - z_{j-1}^2}{2} = \frac{z_N^2 - z_0^2}{2}$$

(6)



The parameterization $z(t) = 3e^{i\theta}$, $0 \leq \theta \leq \pi$.

$$\begin{aligned}
 \int_C \operatorname{Log}(z) dz &= \int_0^\pi \operatorname{Log}(3e^{i\theta}) \cdot 3ie^{i\theta} d\theta \\
 &= \int_0^\pi (\ln|3| + i\theta) 3ie^{i\theta} d\theta \\
 &= 3i \ln 3 \int_0^\pi e^{i\theta} d\theta - 3 \int_0^\pi \theta e^{i\theta} d\theta \\
 &= 3i \ln 3 \cdot \frac{2}{i} - 3(i\pi - 2) \\
 &= 6 \ln 3 + 6 - 3\pi i
 \end{aligned}$$

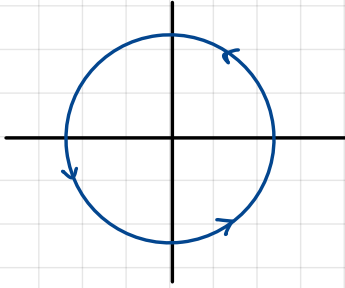


The parameterization $z(t) = 3e^{-i\theta}$, $0 \leq \theta \leq \pi$.

$$\begin{aligned}
 \int_C \operatorname{Log}(z) dz &= \int_0^\pi \operatorname{Log}(3e^{-i\theta}) \cdot (-3ie^{-i\theta}) d\theta \\
 &= \int_0^\pi (\ln|3| - i\theta) (-3ie^{-i\theta}) d\theta \\
 &= -3i \ln|3| \int_0^\pi e^{-i\theta} d\theta - 3 \int_0^\pi \theta e^{-i\theta} d\theta \\
 &= -3i \ln 3 \cdot \left. -\frac{1}{i} e^{-i\theta} \right|_0^\pi - 3(-\pi i - 2) \\
 &= 3 \ln 3 (-1 - 1) + 6\pi i + 6 \\
 &= 6 \ln 3 + 3\pi i + 6.
 \end{aligned}$$

$$\begin{aligned}
 \left(\int_0^\pi \theta e^{i\theta} d\theta \right. &= \left. \frac{1}{i} \theta e^{i\theta} \Big|_0^\pi - \frac{1}{i} \int_0^\pi e^{i\theta} d\theta \right. \\
 &= \left. \frac{1}{i} \pi e^{i\pi} - \frac{1}{i} \frac{1}{i} e^{i\theta} \Big|_0^\pi \right) \\
 &= i\pi - 2
 \end{aligned}$$

So the integral around the full circle is:



$$\begin{aligned}
 \int_C \operatorname{Log}(z) dz &= \int_{-C_d} \operatorname{Log}(z) dz + \int_{C_u} \operatorname{Log}(z) dz \\
 &= -(6 \ln 3 + 3\pi i + 6) + \\
 &\quad 6 \ln 3 + 6 - 3\pi i \\
 &= -6\pi i.
 \end{aligned}$$

Upper Bounds for Moduli

Being able to put bounds on the value of integrals will be very useful to us later on. One way to do this is to take the modulus.

Lemma: If $w(t)$ is a piecewise continuous complex-valued function defined on an interval $a \leq t \leq b$, then

$$\left| \int_a^b w(t) dt \right| \leq \int_a^b |w(t)| dt$$

Proof: Let $\int_a^b w(t) dt =: r_0 e^{i\theta_0}$

$$\Rightarrow r_0 = \int_a^b w(t) e^{-i\theta_0} dt \quad \text{is a real number.}$$

$$\begin{aligned}
 \Rightarrow r_0 &= \operatorname{Re} \int_a^b w(t) e^{-i\theta_0} dt \\
 &= \int_a^b \operatorname{Re}(w(t) e^{-i\theta_0}) dt
 \end{aligned}$$

$$\text{But,} \quad \operatorname{Re}(w(t) e^{i\theta_0}) \leq |w(t) e^{i\theta_0}| = |w(t)|$$

$$\Rightarrow r_0 \leq \int_a^b |w(t)| dt.$$

□

For a contour integral, we get a similar result but need to account for the length of the path.

Thm: Let C be a contour of length L and suppose $f(z)$ is piecewise continuous on C . If M is a non-negative constant s.t. $|f(z)| \leq M$ for all $z \in C$, then

$$\left| \int_C f(z) dz \right| \leq ML.$$

Proof. Let $z(t)$ be a parameterization of C . Then by the lemma above:

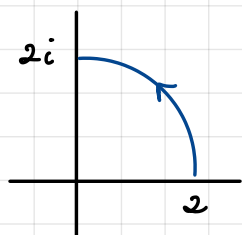
$$\begin{aligned} \left| \int_C f(z) dz \right| &= \left| \int_a^b f(z(t)) z'(t) dt \right| \leq \int_a^b |f(z(t))| |z'(t)| dt \\ &\leq M \int_a^b |z'(t)| dt = ML \quad \square \end{aligned}$$

Remark: We say f is piecewise continuous on the contour C if $f(z(t))$ is piecewise continuous at all but a finite number of values for t and that at these points $f(z(t))$ has one-sided limits.

Since f is continuous on each of a finite union of closed intervals, $|f|$ must achieve a maximum value. That is, we can always find such a value M and can take it to be

$$M = \sup_{z \in C} |f(z)|.$$

Example. How can we bound: $\left| \int_C \frac{z+4}{z^3-1} dz \right|$?



Using the triangle inequality:

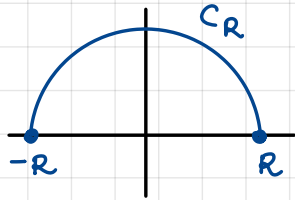
$$\left| \frac{z+4}{z^3-1} \right| = \left| \frac{2e^{i\theta} + 4}{2^3 e^{i3\theta} - 1} \right| \leq \frac{|2| + 4}{||2^3| - 1|} = \frac{6}{7}$$

(Recall the reverse triangle inequality: $|a-b| \geq ||a| - |b||$.)

Since the length of the path is $\frac{2\pi \cdot 2}{4} = \pi$

$$\Rightarrow \left| \int_C \frac{z+4}{z^3-1} dz \right| \leq \frac{6\pi}{7}.$$

Example:



Suppose $z^{1/2}$ denotes the principal value of the square root.

Then what is $\lim_{R \rightarrow \infty} \int_{C_R} \frac{z^{1/2}}{z^2+1} dz$?

$$\begin{aligned} \text{We know } z^{1/2} &= \exp\left(\frac{1}{2} \log z\right) = \exp\left(\frac{1}{2} \log(Re^{i\theta})\right) \\ &= \exp\left(\ln R^{1/2} + \frac{1}{2} i\theta\right) \\ &= \sqrt{R} e^{i\theta/2} \end{aligned}$$

$$\Rightarrow |z^{1/2}| = \sqrt{R}.$$

$$\text{Also } |z^2+1| \geq ||z^2|-1| = R^2-1.$$

$$\text{Hence, } \lim_{R \rightarrow \infty} \left| \int_{C_R} \frac{z^{1/2}}{z^2+1} dz \right| \leq \lim_{R \rightarrow \infty} \frac{\sqrt{R}}{R^2-1} \cdot \pi R = 0.$$

$$\text{We conclude that } \lim_{R \rightarrow \infty} \int_{C_R} \frac{z^{1/2}}{z^2+1} dz = 0.$$

Antiderivatives

We have now seen examples of contour integrals that seem to be independent of the path taken between the end points. In particular, this means that some contour integrals around closed loops are zero.

It turns out this is related to the existence of an anti-derivative of the function f . Note that an anti-derivative is necessarily analytic and so we have a fundamental theorem of calculus.

Thm: Suppose $f(z)$ is continuous on a domain D .
The following statements are equivalent:

- (a) $f(z)$ has an anti-derivative $F(z)$ on D
- (b) the integrals of $f(z)$ along contours $C \subseteq D$ with end points z_1 and z_2 all have the same value
- $$\int_C f(z) dz = F(z_2) - F(z_1).$$

where $F(z)$ is the anti-derivative from (a)

- (c) the integrals of $f(z)$ around a closed contour lying entirely inside D are zero.

When this theorem applies we sometimes write

$$\int_C f(z) dz = \int_{z_1}^{z_2} f(z) dz$$

Examples:

- (1) The function $f(z) = z^2$ has the anti-derivative

$$F(z) = \frac{z^3}{3} \text{ by our differentiation rules.}$$

$$\int_0^{1+i} z^2 dz = \left. \frac{z^3}{3} \right|_0^{1+i} = \frac{1}{3} (1+i)^3 = \frac{2}{3} (-1+i)$$

- (2) The function $f(z) = \frac{1}{z^2}$ which is continuous everywhere except at the origin, has an anti-derivative

$$F(z) = -\frac{1}{z} \text{ for } |z| > 0.$$

$$\Rightarrow \int_C \frac{dz}{z^2} = 0 \quad \text{when } C \text{ is the circle of radius 2 centered at } 0.$$

Proof: "(a) $\Rightarrow$ (b)" Suppose $F(z)$ is the anti-derivative for $f(z)$ and $z(t)$ parameterizes the contour C .

Then if $z(t)$ is a single smooth arc:

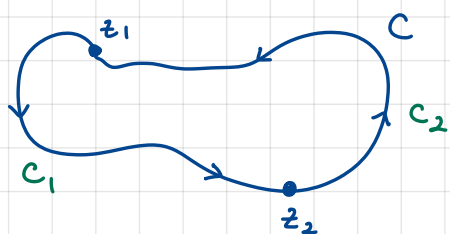
$$\begin{aligned}\int_C f(z) dz &= \int_a^b f(z(t)) z'(t) dt \\ &= \int_a^b \frac{d}{dt} (F(z(t))) dt = F(z_2) - F(z_1).\end{aligned}$$

If C is a contour made up of a finite number of smooth arcs z_j for $j=1, \dots, N$ with endpoints w_{j-1}, w_j (where $w_0 = z_1, w_N = z_2$) then

$$\begin{aligned}\int_C f(z) dz &= \sum_{j=1}^N \int_a^b f(z_j(t)) z_j'(t) dt \\ &= \sum_{j=1}^N F(w_j) - F(w_{j-1}) = F(w_N) - F(w_0) \\ &= F(z_2) - F(z_1).\end{aligned}$$

"(b) $\Rightarrow$ (c)" Suppose that for any contour C with end points z_1 and z_2 the integral has the same value.

If C is a closed contour inside D , then we can break it into two pieces at any two distinct points z_1 and z_2 .



We call these contours C_1 and C_2 and note by assumption that

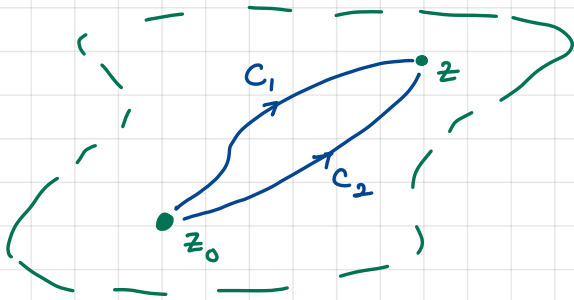
$$\begin{aligned}\int_C f(z) dz &= \int_{C_1 + C_2} f(z) dz \\ &= \int_{C_1} f(z) dz - \int_{-C_2} f(z) dz = 0\end{aligned}$$

since C_1 and $-C_2$ have the same endpoints.

"(c) $\Rightarrow$ (a)" Suppose that for any closed contour C in D the contour integral has value zero.

We want to use this to find an anti-derivative for f .

We do this by using a contour joining a reference point z_0 and the evaluation point z .



Since D is a domain, we can find a contour joining z to z_0 made up of finitely many straightlines. Call this contour C_1 .

We define $F(z) := \int_{C_1} f(w)dw$

First we want to show F is independent of the choice of contour. If C_2 is another contour joining z_0 and z then by assumption:

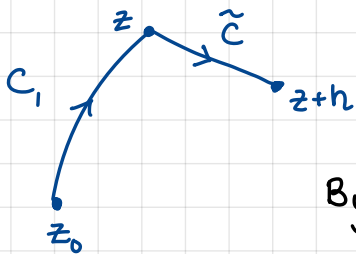
$$0 = \int_{C_1 - C_2} f(w)dw = \int_{C_1} f(w)dw + \int_{-C_2} f(w)dw$$

$$\Rightarrow \int_{C_2} f(w)dw = \int_{C_1} f(w)dw$$

Second we verify that this is an anti-derivative. By definition the derivative of F is given by the limit (when it exists):

$$\frac{dF}{dz}(z) = \lim_{h \rightarrow 0} \frac{F(z+h) - F(z)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left(\int_{z_0}^{z+h} f(w)dw - \int_{z_0}^z f(w)dw \right)$$



By joining z to $z+h$ by a contour we can find

$$\int_{z_0}^{z+h} f(w)dw = \int_{z_0}^z f(w)dw + \int_z^{z+h} f(w)dw$$

$$\Rightarrow \frac{dF}{dz}(z) = \lim_{h \rightarrow 0} \left(\frac{1}{h} \int_z^{z+h} f(w)dw \right)$$

To show the limit exists we can use the ε - δ definition and the fact that we expect the limit to be $f(z)$.

Note that $\int_z^{z+h} f(z)dw = f(z) \int_z^{z+h} 1dw = hf(z)$

since the function f has an anti-derivative its contour integral is independent of the path (by "(a) $\Rightarrow$ (b)").

$$\begin{aligned} \frac{1}{h} \int_z^{z+h} f(w) dw - f(z) &= \frac{1}{h} \int_z^{z+h} f(w) dw - \frac{1}{h} \int_z^{z+h} f(z) dw \\ &= \frac{1}{h} \int_z^{z+h} f(w) - f(z) dw \end{aligned}$$

Now by continuity of f , for any ε we know that

$$|f(w) - f(z)| < \varepsilon \quad \text{whenever} \quad |w - z| < \delta.$$

So choosing $|h|$ small enough we have $w \in \tilde{C}$, the contour joining z and $z+h$ which is inside the ball $\{|w - z| < \delta\}$ and hence

$$\begin{aligned} \left| \frac{1}{h} \int_z^{z+h} f(w) dw - f(z) \right| &\leq \frac{1}{h} \int_z^{z+h} |f(w) - f(z)| dy \\ &\leq \frac{1}{h} \int_z^{z+h} \varepsilon dy = \varepsilon. \end{aligned}$$

We conclude the limit exists and $\frac{dF(z)}{dz} = f(z)$ □

To summarize: we know that if

$\int_C f(z) dz$ depends only on the endpoints of C , for all contours C in a domain D	$\Leftrightarrow$	$\exists$ anti-derivative F for f in D ie. $\frac{d}{dz} (F(z)) = f(z)$ for all $z \in D$.
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Cauchy - Goursat Theorem

We can find a second criteria for the path-independence of contour integrals in terms of the analyticity of the function f .

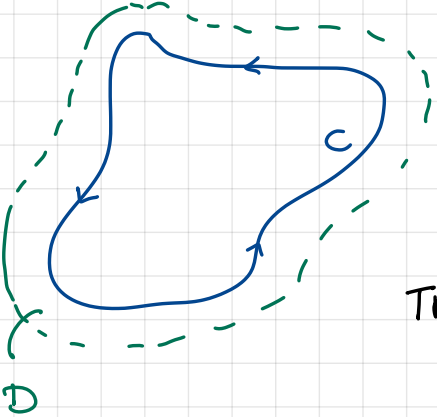
Let C be a simple closed contour oriented positively, and let f be an analytic function on a domain containing the interior of C and the curve C itself.

f is analytic on D and so

$$f(z) = u(x,y) + iv(x,y)$$

and u, v satisfy the Cauchy-Riemann equations.

The contour integral of f can be written in terms of u and v :


$$\begin{aligned}\int_C f(z) dz &= \int_a^b f(z(t)) z'(t) dt \\&= \int_a^b (u(x(t), y(t)) + iv(x(t), y(t))) \cdot (x'(t) + iy'(t)) dt \\&= \int_a^b u(x(t), y(t)) x'(t) - v(x(t), y(t)) y'(t) dt \\&\quad + i \int_a^b u(x(t), y(t)) y'(t) + v(x(t), y(t)) x'(t) dt \\&= \int_a^b u x' - v y' dt + i \int_a^b u y' + v x' dt \\&= \int_C u dx - v dy + i \int_C u dy + v dx\end{aligned}$$

This expression is a differential form, and can be evaluated using Green's Theorem.

In Vector Calculus, Green's Theorem in 2D describes a relationship between the integral over a closed region and its boundary.

Thm: (Green's Theorem 2D)

Suppose L, M are functions of x, y with continuous partial derivatives. Moreover, let C be a closed curve containing a region $\tilde{D}$. Then

anticlockwise
integral $\rightarrow \oint_C L dx + M dy = \iint_{\tilde{D}} \frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} dx dy.$

Applying Green's Theorem to the above:

$$\begin{aligned} \int_C f(z) dz &= \iint_R -\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} dx dy + i \iint_R -\frac{\partial v}{\partial y} + \frac{\partial u}{\partial x} dx dy \\ &= 0 \quad \text{by the Cauchy-Riemann equations.} \end{aligned}$$

However note this argument requires that f be analytic and f' be continuous to apply Green's Theorem.

In fact, using complex analysis we can remove the assumption that f' needs to be continuous. This is important, as we will then be able to prove that the derivative of an analytic function is always continuous (and in fact analytic) using this result.

Thm: Cauchy-Goursat

If f is analytic on a domain D containing a simple closed contour C and its interior then

$$\int_C f(z) dz = 0.$$